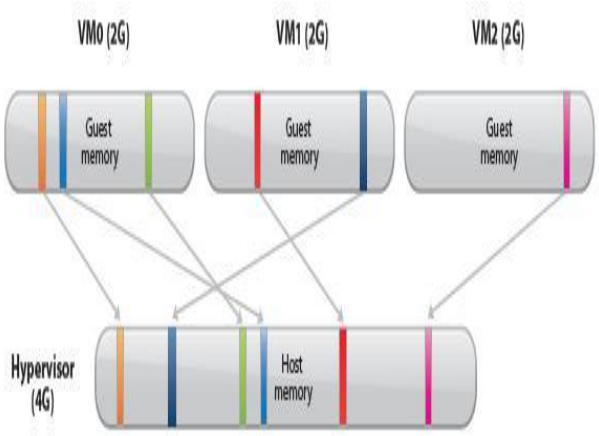


CLOUD COMPUTING COURSE

Lecture27- Virtualization Features(2)

ABSTRACT

Lecture27 Introduces virtualization cluster features.

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Virtualization Features

Virtualization Cluster Features

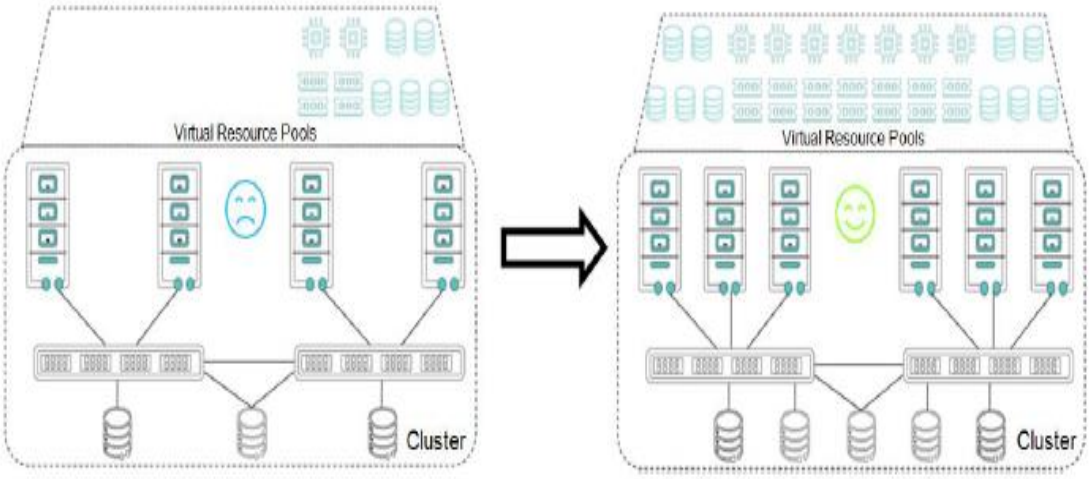
3. Scalability

In a traditional non-virtualization environment, all services are deployed on physical machines. In the initial phase of system construction, the service volume is not large. Therefore, the hardware resources configured for physical machines are limited. As the service volume increases, the original hardware cannot meet service requirements and needs to be upgraded continuously, for example, adding CPUs or expanding memory from 256 GB to 512 GB. This **scaling mode is called vertical scaling, or scale-up.**

However, the hardware resources supported by a physical machine have an upper limit. If the service volume keeps increasing, the physical machine must be replaced, and the service migration requires machine shutdown.

In virtualization, all resources are pooled to carry service VMs. When the service volume increases, you only need to add resources to the resource pool, without having to add hardware resources. During the actual implementation, you only need to add servers. This scaling mode is called **horizontal scaling or scale-out.**

Clusters support horizontal scaling, which is easier than the scaling in traditional non-virtualized IT systems.



Scaling in the Cluster

As shown in the figure, the original cluster contains only four servers and three storage devices. At the early stage of construction, the resources in the resource pool are sufficient. After some resources are consumed by VMs, the administrator finds that the existing resources are insufficient to support new services.

1. The administrator can migrate all VMs to certain physical servers in a cluster during off-peak hours and power off and upgrade other hardware resources. (Scale-up)

2. The administrator can also add servers and storage devices to a cluster without affecting services. (Scale-out)

Scale-up or scale-out in virtualization systems is easier to implement than scaling in non-virtualization systems and has less impact on services.

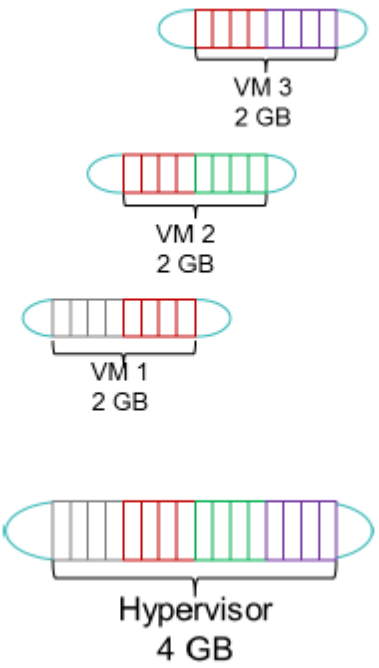
4. Memory Overcommitment

Memory overcommitment allows VMs to use more memory than the total physical memory of the server by leveraging specific technologies to improve VM density and release free memory when the memory is insufficient.

Memory overcommitment technologies include: **memory ballooning**, **memory swapping**, and **memory sharing**.

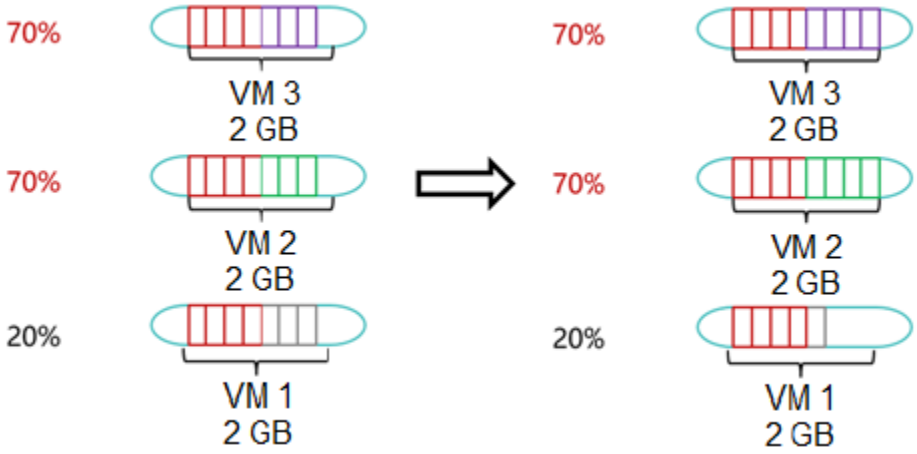
Generally, **the three technologies need to be applied <u>together</u>** and take effect at the same time.

1. **Memory sharing**: Multiple VMs share the memory page on which the data content is the same.



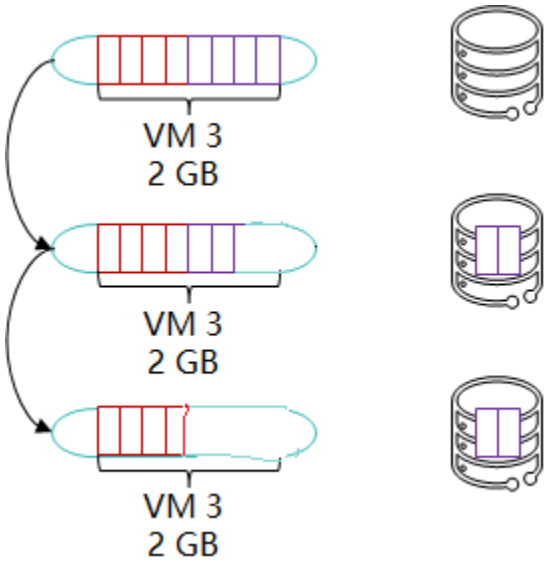
Memory Sharing

2. **Memory Ballooning**: The system automatically reclaims the unused memory from a VM and allocates it to other VMs to use.
   Applications on the VMs are not aware of memory reclamation and allocation. The total amount of the memory **used** by all VMs on a physical server cannot exceed the physical memory of the server.



Memory Ballooning

3. **Memory Swapping**: External storage is virtualized into memory for VMs to use. Data that is not used temporarily is stored to external storage. If the data needs to be used, it is exchanged with the data reserved on the memory.



Memory Swapping